

Heat Transfer Operations – I

T2430C3

Textbooks:

1. Cengel A. Yunnus. *“Heat Transfer – A Practical Approach”*, McGraw Hill.
 2. McCabe W. L., Smith J. C., Harriot P., Unit Operations of Chemical Engineering, 5th edition, McGraw Hill, 1993.
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1. Binay. K. Dutta, *“Heat Transfer Principles and applications”* Prentice Hall of India.
 2. Kern D Q, *“Process Heat Transfer”*, McGraw Hill Book Co. (1997).

Unit 1 – Introduction - Cengel

- Overview of applications of heat transfer in different fields of engineering
- Modes of heat transfer- conduction, convection and radiation
- Dimensionless numbers
- Heat transfer with and without change of phase

Modes of heat transfer- conduction, convection and radiation

- **Heat is the form of energy that can be transferred from one system to another as a result of temperature difference.**
- The science that deals with the **determination of the rates of such energy transfers** is heat transfer.
- Three basic mechanisms of heat transfer, which are conduction, convection, and radiation.
- **Conduction** is the transfer of energy from the more energetic particles of a substance to the adjacent, less energetic ones as a result of interactions between the particles.
- **Convection** is the mode of heat transfer between a solid surface and the adjacent liquid or gas that is in motion, and it involves the combined effects of conduction and fluid motion.
- **Radiation** is the energy emitted by matter in the form of electromagnetic waves (or photons) as a result of the changes in the electronic configurations of the atoms or molecules.

Modes of heat transfer- conduction, convection and radiation

- **The basic requirement for heat transfer is the presence of a temperature difference.**
- There can be no net heat transfer between two bodies that are at the same temperature.
- The rate of heat transfer in a certain direction depends on the magnitude of the temperature gradient (the temperature difference per unit length or the rate of change of temperature) in that direction.
- The larger the temperature gradient, the higher the rate of heat transfer.

Thermodynamics	Heat Transfer
Thermodynamics is concerned with the amount of heat transfer as a system undergoes a process from one equilibrium state to another, and it gives no indication about how long the process will take	Heat is the form of energy that can be transferred from one system to another as a result of temperature difference. The science that deals with the determination of the rates of such energy transfers is heat transfer.
A thermodynamic analysis simply tells us how much heat must be transferred to realize a specified change of state to satisfy the conservation of energy principle.	In practice we are more concerned about the rate of heat transfer (heat transfer per unit time) than we are with the amount of it.
Thermodynamics deals with equilibrium states and changes from one equilibrium state to another.	Heat transfer deals with systems that lack thermal equilibrium, and thus it is a nonequilibrium phenomenon.
	The study of heat transfer cannot be based on the principles of thermodynamics alone. However, the laws of thermodynamics lay the framework for the science of heat transfer.
	The first law requires that the rate of energy transfer into a system be equal to the rate of increase of the energy of that system. The second law requires that heat be transferred in the direction of decreasing temperature

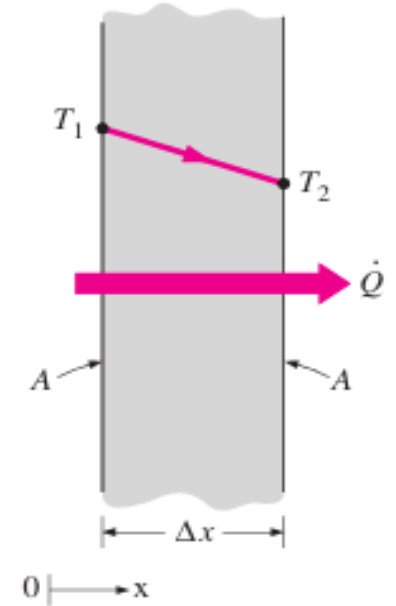
Conduction

- Conduction is the transfer of energy from the more energetic particles of a substance to the adjacent less energetic ones as a result of interactions between the particles.
- In gases and liquids, conduction is due to the collisions and diffusion of the molecules during their random motion.
- In solids, it is due to the combination of vibrations of the molecules in a lattice and the energy transport by free electrons.
- **The rate of heat conduction through a medium depends on the geometry of the medium, its thickness, and the material of the medium, as well as the temperature difference across the medium.**
- ***The rate of heat conduction through a plane layer is proportional to the temperature difference across the layer and the heat transfer area, but is inversely proportional to the thickness of the layer.***

Conduction

Rate of heat conduction $\propto \frac{(\text{Area})(\text{Temperature difference})}{\text{Thickness}}$

$$\dot{Q}_{\text{cond}} = kA \frac{T_1 - T_2}{\Delta x} = -kA \frac{\Delta T}{\Delta x}$$



- where the **constant of proportionality k is the thermal conductivity of the material**, which is a measure of the ability of a material to conduct heat.
- In the limiting case of $\Delta x \rightarrow 0$, the equation above reduces to the differential form which is called **Fourier's law of heat conduction after J. Fourier**, who expressed it first in his heat transfer text in 1822.
- Here **dT/dx is the temperature gradient**, which is the slope of the temperature curve on a T-x diagram (the rate of change of T with x), at location x.

Conduction

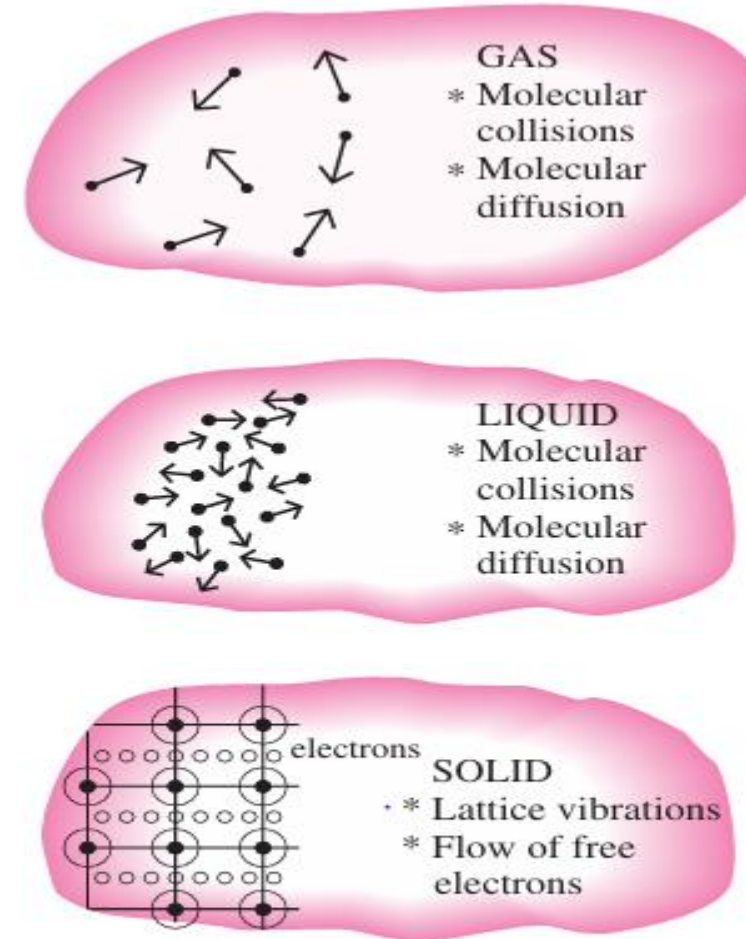
- Heat is conducted in the direction of decreasing temperature, and the temperature gradient becomes negative when temperature decreases with increasing x .
- The negative sign in Eq. ensures that heat transfer in the positive x direction is a positive quantity.
- **The heat transfer area A is always normal to the direction of heat transfer.**
- The thermal conductivity k is a measure of a material's ability to conduct heat.
- **The thermal conductivity of a material can be defined as the rate of heat transfer through a unit thickness of the material per unit area per unit temperature difference.**
- A high value for thermal conductivity indicates that the material is a good heat **conductor**, and a low value indicates that the material is a poor heat conductor or **insulator**.

TABLE 1–1

The thermal conductivities of some materials at room temperature

Material	k, W/m · °C*
Diamond	2300
Silver	429
Copper	401
Gold	317
Aluminum	237
Iron	80.2
Mercury (l)	8.54
Glass	0.78
Brick	0.72
Water (l)	0.613
Human skin	0.37
Wood (oak)	0.17
Helium (g)	0.152
Soft rubber	0.13
Glass fiber	0.043
Air (g)	0.026
Urethane, rigid foam	0.026

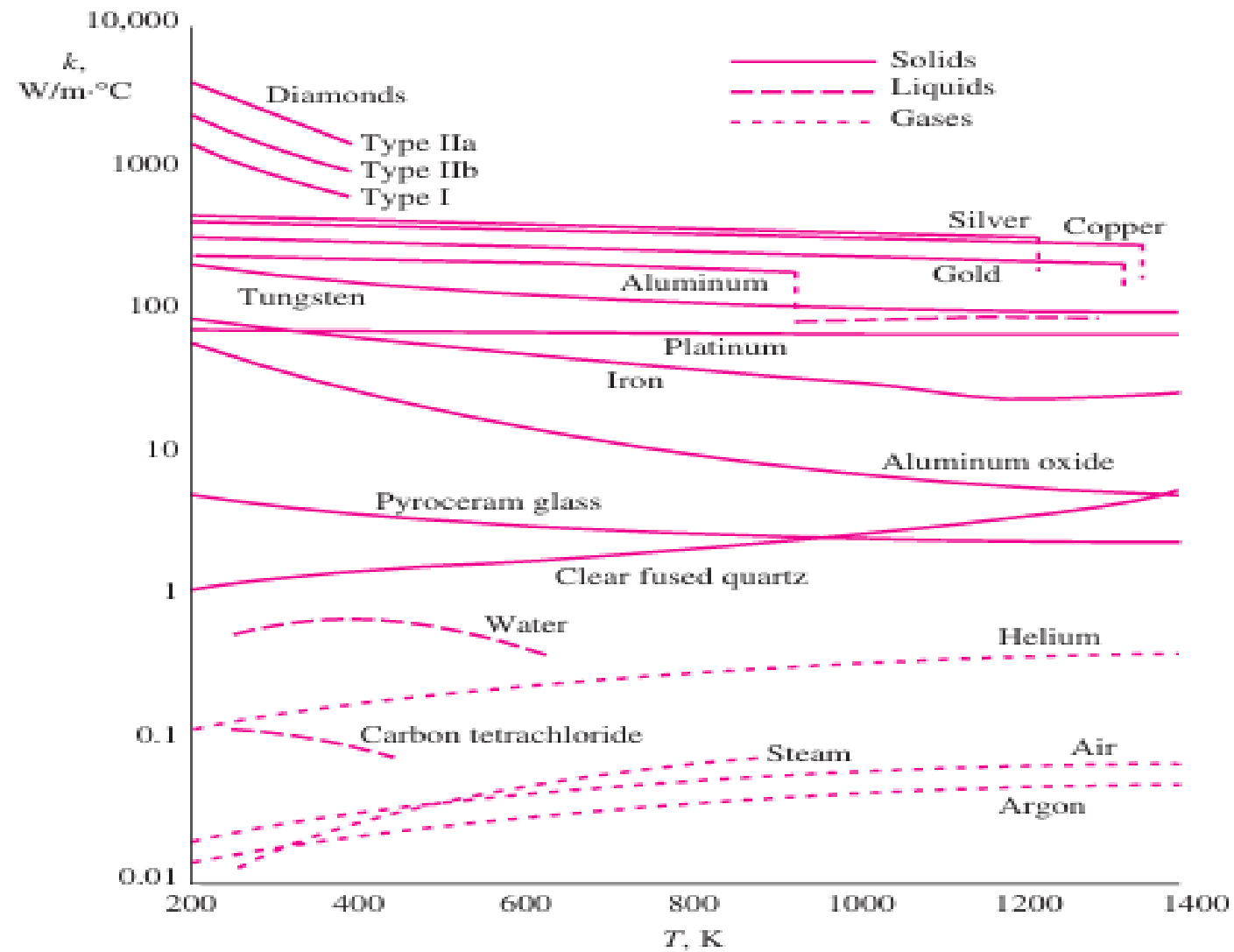
*Multiply by 0.5778 to convert to Btu/h · ft · °F.

**FIGURE 1–27**

The mechanisms of heat conduction in different phases of a substance.

Conduction

- **The thermal conductivity of a gas increases with increasing temperature and decreasing molar mass (kinetic theory).**
- **The thermal conductivities of most liquids decrease with increasing temperature**, with water being a notable exception, the conductivity of liquids decreases with increasing molar mass.
- Liquid metals such as mercury and sodium have high thermal conductivities and are very suitable for use in applications where a high heat transfer rate.
- The relatively high thermal conductivities of pure metals are primarily due to the electronic component. The lattice component of thermal conductivity strongly depends on the way the molecules are arranged.
- Unlike metals, which are good electrical and heat conductors, crystalline solids such as diamond and semiconductors such as silicon are good heat conductors but poor electrical conductors.
- Pure metals have high thermal conductivities but decrease with increase in temperature, thermal conductivity of an alloy of two metals is usually much lower than that of either metal.
- The thermal conductivities of certain solids exhibit dramatic increases at temperatures near absolute zero, when these solids become superconductors.



Conduction

- **Thermal diffusivity** represents how fast heat diffuses through a **material** and is defined as

$$\alpha = \frac{\text{Heat conducted}}{\text{Heat stored}} = \frac{k}{\rho C_p} \quad (\text{m}^2/\text{s})$$

- **The thermal conductivity k represents how well a material conducts heat**, and the heat capacity C_p represents how much energy a material stores per unit volume.
- The thermal diffusivity of a material can be viewed as the ratio of the heat conducted through the material to the heat stored per unit volume.
- A material that has a high thermal conductivity or a low heat capacity will obviously have a large thermal diffusivity.
- The larger the thermal diffusivity, the faster the propagation of heat into the medium.

The thermal diffusivities of some materials at room temperature

Material	$\alpha, \text{m}^2/\text{s}^*$
Silver	149×10^{-6}
Gold	127×10^{-6}
Copper	113×10^{-6}
Aluminum	97.5×10^{-6}
Iron	22.8×10^{-6}
Mercury (l)	4.7×10^{-6}
Marble	1.2×10^{-6}
Ice	1.2×10^{-6}
Concrete	0.75×10^{-6}
Brick	0.52×10^{-6}
Heavy soil (dry)	0.52×10^{-6}
Glass	0.34×10^{-6}
Glass wool	0.23×10^{-6}
Water (l)	0.14×10^{-6}
Beef	0.14×10^{-6}
Wood (oak)	0.13×10^{-6}

Convection

- **Convection is the mode of energy transfer between a solid surface and the adjacent liquid or gas that is in motion, and it involves the combined effects of conduction and fluid motion.**
- The faster the fluid motion, the greater the convection heat transfer.
- In the absence of any bulk fluid motion, heat transfer between a solid surface and the adjacent fluid is by pure conduction.
- The presence of bulk motion of the fluid enhances the heat transfer between the solid surface and the fluid, but it also complicates the determination of heat transfer rates.
- Convection is called **forced convection** if the fluid is forced to flow over the surface by external means such as a fan, pump, or the wind.
- Convection is called **natural (or free) convection** if the fluid motion is caused by buoyancy forces that are induced by density differences due to the variation of temperature in the fluid

Convection

- Heat transfer processes that involve change of phase of a fluid are also considered to be convection because of the fluid motion induced during the process, such as the rise of the vapor bubbles during boiling or the fall of the liquid droplets during condensation.
- Despite the complexity of convection, the rate of convection heat transfer is observed to be proportional to the temperature difference, and is conveniently expressed by **Newton's law of cooling** as

$$\dot{Q}_{\text{conv}} = hA_s (T_s - T_{\infty}) \quad (\text{W})$$

- where h is the convection heat transfer coefficient in $\text{W/m}^2 \cdot ^\circ\text{C}$
- A_s is the surface area through which convection heat transfer takes place, T_s is the surface temperature, and T_{∞} is the temperature of the fluid sufficiently far from the surface.

Convection

- The convection heat transfer coefficient h is not a property of the fluid.
- **h is an experimentally determined parameter whose value depends on all the variables influencing convection such as the surface geometry, the nature of fluid motion, the properties of the fluid, and the bulk fluid velocity.**
- *Convection = conduction with fluid motion (Essentially)*

Typical values of convection heat transfer coefficient

Type of convection	$h, \text{W/m}^2 \cdot ^\circ\text{C}^*$
Free convection of gases	2–25
Free convection of liquids	10–1000
Forced convection of gases	25–250
Forced convection of liquids	50–20,000
Boiling and condensation	2500–100,000

Radiation

- **Radiation is the energy emitted by matter in the form of electromagnetic waves (or photons) as a result of the changes in the electronic configurations of the atoms or molecules.**
- Unlike conduction and convection, the transfer of energy by radiation does not require the presence of an intervening medium.
- Energy transfer by radiation is fastest (at the speed of light) and it suffers no attenuation in a vacuum.
- This is how the energy of the sun reaches the earth.
- In heat transfer studies, interest is in thermal radiation, which is the form of radiation emitted by bodies because of their temperature.
- All bodies at a temperature above absolute zero emit thermal radiation.
- Radiation is a volumetric phenomenon, and all solids, liquids, and gases emit, absorb, or transmit radiation to varying degrees.

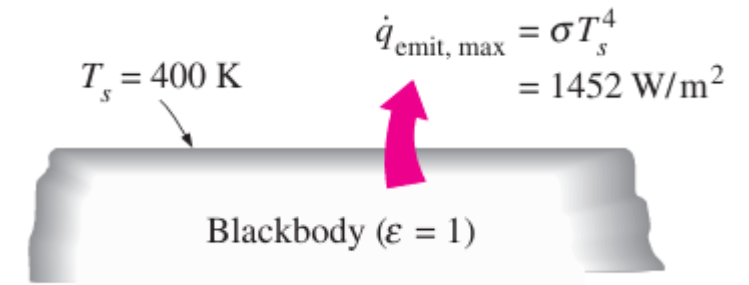
Radiation

- Radiation is usually considered to be a surface phenomenon for solids that are opaque to thermal radiation such as metals, wood, and rocks.
- Since the radiation emitted by the interior regions of such material can never reach the surface, and the radiation incident on such bodies is usually absorbed within a few microns from the surface.
- The maximum rate of radiation that can be emitted from a surface at an absolute temperature T_s (in K or R) is given by the **Stefan–Boltzmann law** as

$$\dot{Q}_{\text{emit, max}} = \sigma A_s T_s^4 \quad (\text{W})$$

- where $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$ is the Stefan–Boltzmann constant.

Radiation



- The idealized surface that emits radiation at this **maximum rate is called a blackbody**, and the radiation emitted by a black body is called blackbody radiation.
- Blackbody radiation represents the maximum amount of radiation that can be emitted from a surface at a specified temperature.
- The radiation emitted by all real surfaces is less than the radiation emitted by a blackbody at the same temperature, and is expressed as

$$\dot{Q}_{\text{emit}} = \epsilon \sigma A_s T_s^4 \quad (\text{W})$$

- Where ϵ is the **emissivity of the surface**.
- The property emissivity, whose value is in the **range $0 \leq \epsilon \leq 1$** , is a measure of how closely a surface approximates a **blackbody for which 1**.

Radiation

- Another important radiation property of a surface is its **absorptivity α** , which is the fraction of the radiation energy incident on a surface that is absorbed by the surface.
- Like emissivity, its value is in the range **$0 \leq \alpha \leq 1$** .
- A blackbody absorbs the entire radiation incident on it. That is, a blackbody is a perfect absorber ($\alpha = 1$) as it is a perfect emitter.
- In general, both ϵ and α of a surface depend on the temperature and the wavelength of the radiation.
- **Kirchhoff's law of radiation states that the emissivity and the absorptivity of a surface at a given temperature and wavelength are equal.**

Radiation

- When a surface of emissivity and surface area A_s at an absolute temperature T_s is completely enclosed by a much larger (or black) surface at absolute temperature T_{surr} separated by a gas (such as air) that does not intervene with radiation, the net rate of radiation heat transfer between these two surfaces is given by

$$\dot{Q}_{\text{rad}} = \epsilon \sigma A_s (T_s^4 - T_{\text{surr}}^4) \quad (\text{W})$$

- Radiation heat transfer to or from a surface surrounded by a gas such as air occurs parallel to conduction (or convection, if there is bulk gas motion) between the surface and the gas.
- The total heat transfer rate to or from a surface by convection and radiation is expressed as

$$\dot{Q}_{\text{total}} = h_{\text{combined}} A_s (T_s - T_{\infty}) \quad (\text{W})$$

Overview of applications of heat transfer in different fields of engineering

- The human body is constantly rejecting heat to its surroundings, and human comfort is closely tied to the rate of this heat rejection.
- Many ordinary household appliances are designed, in whole or in part, by using the principles of heat transfer.
- Some examples include the electric or gas range, the heating and air-conditioning system, the refrigerator and freezer, the water heater, the iron, and even the computer, the TV, and the VCR.
- Of course, energy-efficient homes are designed on the basis of minimizing heat loss in winter and heat gain in summer.
- Heat transfer plays a major role in the design of many other devices, such as car radiators, solar collectors, various components of power plants, and even spacecraft.
- The optimal insulation thickness in the walls and roofs of the houses, on hot water or steam pipes, or on water heaters is again determined on the basis of a heat transfer analysis with economic consideration

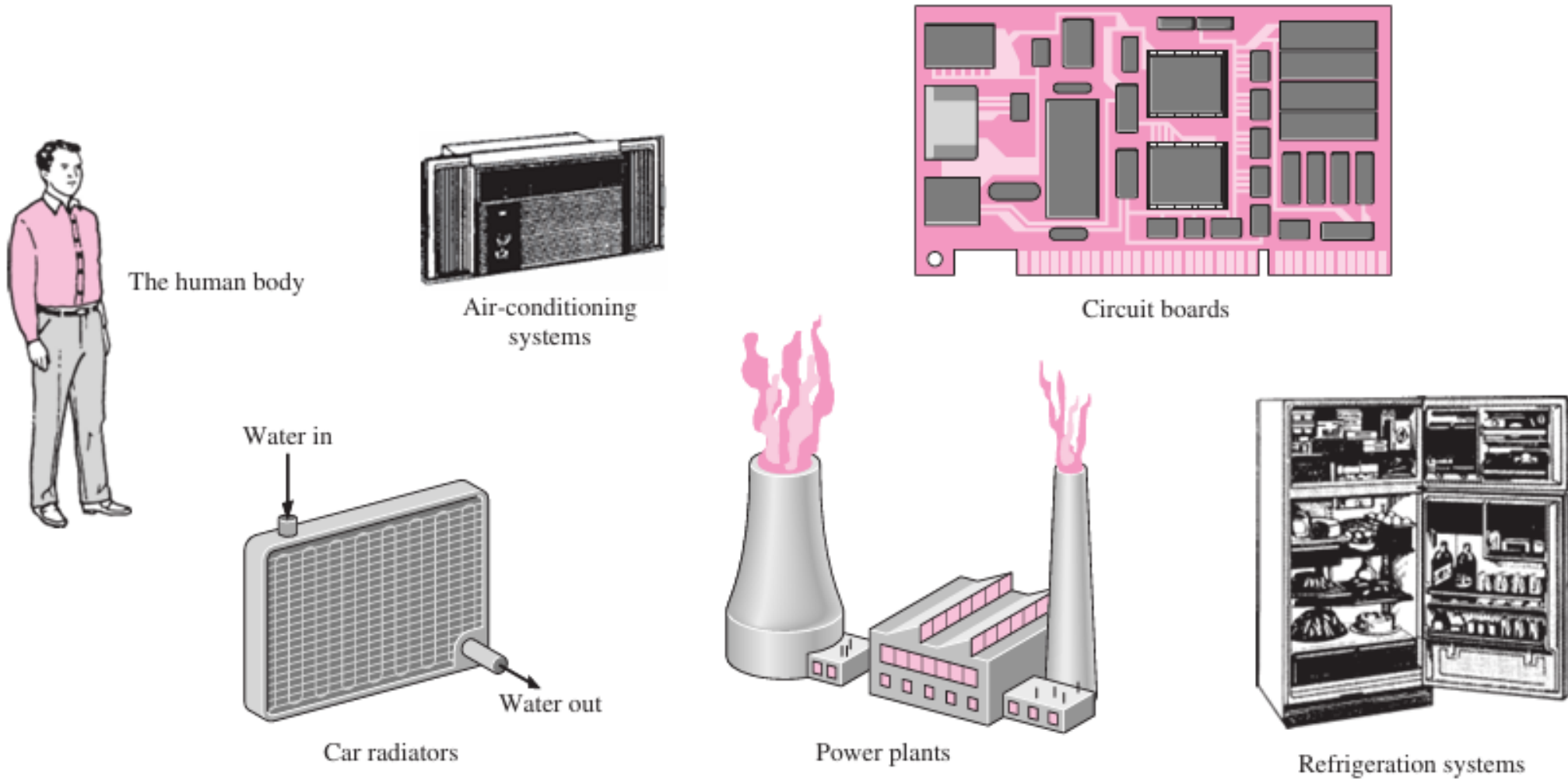


FIGURE 1-3
Some application areas of heat transfer.

Overview of applications of heat transfer in different fields of engineering

Industries that use process heating:

- **Oil & Gas Industry:** Offshore as well as onshore drilling, downhole heating, hydraulic fracturing, and maintaining petrochemicals' viscosity.
- **Packaging & Plastics:** Heaters are used in form-fill-seal machines, portion packaging, tamper-proof packaging, extrusion, and softening polymers to reshape them.
- **Power Plants:** Process air heaters for inlet housing, superheating, sulphur reduction, catalytic cracking and regeneration, gas dehydration, etc.
- **Aerospace:** Heating systems are used to heat air in turbine systems, batteries, engines, cockpits, etc.
- **Food and beverage:** Applications include fryers, ovens, grills, deep fryers, hot plates, and steam tables.

Overview of applications of heat transfer in different fields of engineering

Industries that use process heating:

- **Medical field:** Heating of blankets, jackets, flasks, monitoring of equipment in laboratories, fluid as well as blood warming, etc.
- **Automobile Manufacturing:** Heaters are used for making automobile parts, such as clutch plates, brake shoes, etc.
- **Building & Construction:** Hot water temperature maintenance; freeze protection; frost prevention; pipe freeze protection.
- **Defense Industry:** Heaters are generally used in fuel preheating processes, temperature maintenance in cold regions, as well as lube oil heating.
- **Rubber Industries:** Heaters deliver the ideal solutions in vacuum pressing, gaskets, boots, and injection molding.

Overview of applications of heat transfer in different fields of engineering

Heat transfer is used in various industries, including:

- Metalworking: heat treatment, welding, and casting
- Plastics processing: injection molding and extrusion
- Food production: cooking, sterilization, and packaging
- Textile manufacturing: dyeing, drying, and finishing

Examples of Conduction in Manufacturing

- Heat treatment of metals: conduction occurs between the metal and the heat treatment equipment
- **Welding:** conduction occurs between the welding torch and the workpiece
- **Casting:** conduction occurs between the molten metal and the mold

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Overview of applications of heat transfer in different fields of engineering

Examples of Convection in Manufacturing

- Cooling systems: convection occurs between the coolant and the equipment or product
- **Drying processes:** convection occurs between the hot air and the product
- **Mixing processes:** convection occurs between the fluid and the mixer

Examples of Radiation in Manufacturing

- **Infrared heating:** radiation is used to heat products or equipment
- **Microwave heating:** radiation is used to heat products or equipment
- **Radiative cooling:** radiation is used to cool products or equipment

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Overview of applications of heat transfer in different fields of engineering

- Heat transfer equipment such as heat exchangers, boilers, condensers, radiators, heaters, furnaces, refrigerators, and solar collectors are designed primarily on the basis of heat transfer analysis.
- The heat transfer problems encountered in practice can be considered in two groups:
(1) rating and (2) sizing problems.
- The rating problems deal with the determination of the heat transfer rate for an existing system at a specified temperature difference.
- The sizing problems deal with the determination of the size of a system in order to transfer heat at a specified rate for a specified temperature difference.

Overview of applications of heat transfer in different fields of engineering

- A heat transfer process or equipment can be studied either experimentally (testing and taking measurements) or analytically (by analysis or calculations).
- The experimental approach has the advantage that we deal with the actual physical system, and the desired quantity is determined by measurement, within the limits of experimental error.
- However, this approach is expensive, time-consuming, and often impractical.
- The analytical approach (including numerical approach) has the advantage that it is fast and inexpensive, but the results obtained are subject to the accuracy of the assumptions and idealizations made in the analysis.

Heat transfer with change of phase

- Processes of heat transfer accompanied by phase change are more complex than simple heat exchange between fluids.
- *A phase change involves the addition or subtraction of considerable quantities of heat at constant or nearly constant temperature.*
- The rate of phase change may be governed by rate of heat transfer, but is often influenced by **rate of nucleation of bubbles, drops, or crystals, and by the behavior of new phase after it is formed.**
- **When a phase change takes place, the temperature on one side is CONSTANT, but the presence of boiling/condensing fluids produces heat transfer.**

McCabe W. L., Smith J. C., Harriot P., Unit Operations of Chemical Engineering, 5th edition, McGraw Hill, 1993.

<https://pages.mtu.edu/~fmorriso/cm310/lectures/2015heatlecture09.pdf>

Heat transfer with change of phase

- Important in evaporation and distillation
- **Large heat transfer coefficient h**
- It's important to know in which regime you operate as each regime has different correlations to calculate heat transfer coefficient

McCabe W. L., Smith J. C., Harriot P., Unit Operations of Chemical Engineering, 5th edition, McGraw Hill, 1993.

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Heat transfer with change of phase

Examples:

- Heat transfer from condensing vapors (dropwise and filmwise condensation)
- Condensation of superheated vapors
- Heat transfer to boiling liquids
- Pool boiling of saturated liquids
- Film boiling
- Subcooled boiling

McCabe W. L., Smith J. C., Harriot P., Unit Operations of Chemical Engineering, 5th edition, McGraw Hill, 1993.

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